

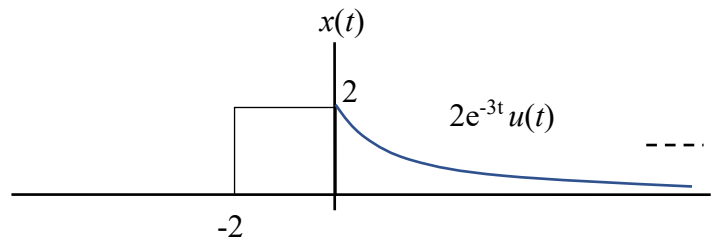
ENGR 3324: Signals and Systems
Sample Test 2

Q1: For the LTIC system described by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

Find the zero-state response $y_{zs}(t)$ of the system if the input is $x(t) = 2e^{-5t}u(t)$. Make sure to use the time-domain analysis

Q2: Use the Laplace integral to Find the Laplace transform $X(s)$ of the signal $x(t)$ shown in Fig. 1 below.



Q3: Find the impulse response $h(t)$ of the system described by the transfer function $H(s)$ below.

$$H(s) = \frac{3s^2 + 2}{s^2 + 3s + 2}$$

Q4: For the LTIC system described by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 9y(t) = \frac{dx(t)}{dt} + x(t)$$

a) Find the transfer function $H(s)$

b) The zero-state response $y_{zs}(t)$ of the system if the input is $x(t) = 3e^{-5t}u(t)$.